

ETHAN (TAEHOON) LEE

US Permanent Resident · ethantlee21@gmail.com · thdxg.dev · linkedin.com/in/ethantlee · github.com/thdxg

EDUCATION

Carnegie Mellon University

Master of Software Engineering - Scalable Systems

Pittsburgh, PA

Expected Dec 2027

Emory University

BS in Computer Science

Atlanta, GA

May 2026

- **Honors:** 3.99/4.0 GPA, Magna Cum Laude, 2x James T Laney Scholarship, 5x Dean's List, Phi Beta Kappa Honor Society

SKILLS

- **Languages:** Go, TypeScript, JavaScript, Java, C, Rust, Python, SQL, Swift, x86 Assembly
- **Technologies:** Kubernetes, Linux, Docker, AWS, GCP, PostgreSQL, Redis, OpenTelemetry, Temporal, Prometheus, Grafana

WORK EXPERIENCE

Founding Software Engineer

Huddle Surety

Atlanta, GA

Sep 2025 – Present

- Developed the core Go REST API spanning 100+ endpoints with Redis caching and Temporal workflow orchestration
- Built multi-stage RAG pipelines with cross-document context retrieval generating bond documents at 95% accuracy
- Shipped a role-based frontend delivering distinct surety agent and contractor UX using Next.js, TypeScript, and tRPC
- Architected an OpenTelemetry observability pipeline with end-to-end context propagation and tracing across services

Software Engineering Intern (Cloud Infrastructure)

UKG

Atlanta, GA

May 2025 – Aug 2025

- Implemented a Kubernetes CronJob that validates TLS certificates of 200+ production MongoDB, MySQL, and RabbitMQ VMs across 4 datacenters, removing manual healthchecks and accelerating the annual certificate renewal process
- Developed a multi-cluster Kubernetes network observability pipeline using OpenTelemetry, Prometheus, and Grafana, scatter-gathering telemetry across 5 clusters behind a unified Go REST API for centralized health monitoring
- Provisioned Prometheus Blackbox Exporters across isolated OpenStack tenants to achieve distributed observability

Software Engineering Intern (Full Stack)

eStreamly

Atlanta, GA

Jun 2024 – Aug 2024

- Optimized global state management and client-side caching on Angular/TypeScript web app, reducing TTFB by 40%
- Created multi-stage container build and release pipeline on GCP Cloud Build, improving deployment speed by 35%

PROJECTS

Kubernetes Homelab & Contribution | *Kubernetes, Linux, FluxCD, Cilium, Grafana, OpenTelemetry* headlamp.thdxg.dev

- Designed a two-node Kubernetes cluster on Raspberry Pi, fully automating CI/CD via GitOps using FluxCD and sops
- Deployed 20+ projects including personal website and Grafana dashboard via Gateway API ingress with dynamic DNS
- Authored the Korean Kubernetes documentation on pod scheduling and eviction as part of SIG Docs localization efforts

DooleyOnline: Emory Student Marketplace | *Go, Next.js, React, TypeScript, PostgreSQL, GCP* github.com/dooleyonline

- Architected a Go REST API spanning item, chat, and user domains with sqlc queries, JWT auth, and email verification
- Implemented real-time buyer-seller chat over WebSockets with persisted message history and S3-backed image uploads

Macterm: MacOS Terminal Emulator with Persistent Multiplexing | *Swift, SwiftUI* github.com/thdxg/macterm

- Built a native macOS terminal with SwiftUI and libhostty, gaining 6K+ downloads and 16 contributors globally
- Implemented session persistence, declarative layout configuration, fuzzy command palette, and configurable hotkeys

Spotify Warped: HCI Research & Publication | *Python, d3.js, React* dl.acm.org/doi/10.1145/3772318.3790875

- Published to ACM CHI 2026 on how Spotify's algorithmic feedback loops shape listening behavior and data engagement
- Built high-throughput data processing pipelines in Python to ingest 100 GB of Spotify streaming data from 60 users